Introduction to Slope-Intercept Form

ANSWER KEY

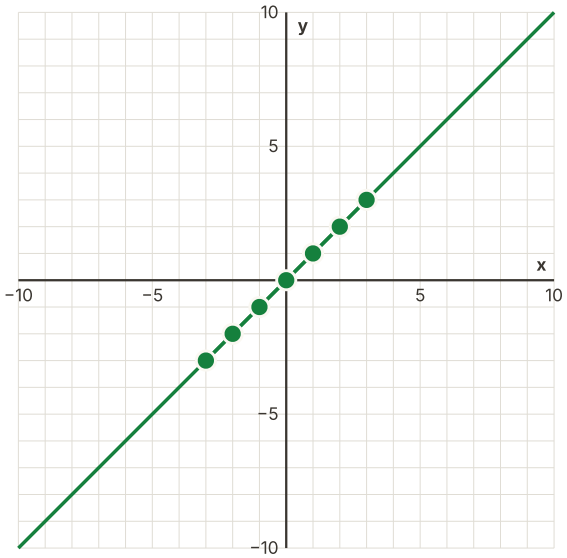
Answer key: every table is filled in and every line is drawn in green.

Words to know

Slope-intercept form: y = mx + b.
Slope (m): how steep the line is. It is the number in front of x.
y-intercept (b): where the line crosses the y-axis, at (0, b).

1 y = x

Input (x)	y = x
−3	−3
−2	−2
−1	−1
0	0
1	1
2	2
3	3



✏ Practice: y = x

1. ★ Try it For y = x, what is y when x = 7?

Answer: 7

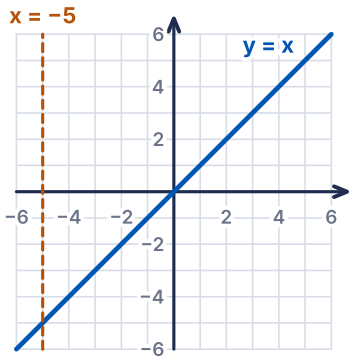
How to do it: y = x, so y = 7.

x	y = x
0	0
1	1
2	2
7	7

2. ★★ On level For y = x, what is y when x = −5?

Answer: −5

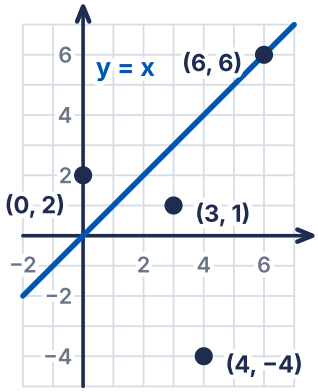
How to do it: y = x, so y = −5. The negative sign stays.



3. ★★★ Challenge Which point is on the line y = x?

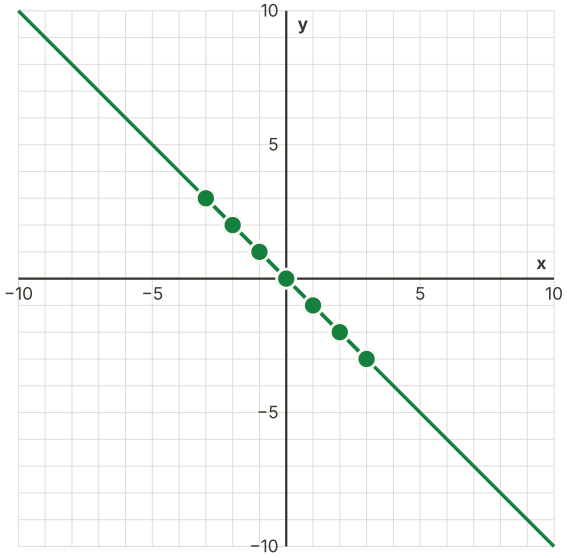
- ○ (4, −4)
- ● **(6, 6)**
- ○ (0, 2)
- ○ (3, 1)

How to do it: On y = x, the x-value and y-value match. Only (6, 6) has matching numbers.



2 y = −x

Input (x)	y = −x
−3	3
−2	2
−1	1
0	0
1	−1
2	−2
3	−3



✏ Practice: y = −x

1. ★ Try it For y = −x, what is y when x = 4?

Answer: −4

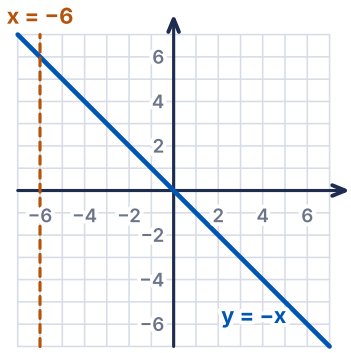
How to do it: y = −(4) = −4.

x	y = −x
0	0
1	−1
2	−2
4	−4

2. ★★ On level For y = −x, what is y when x = −6?

Answer: 6

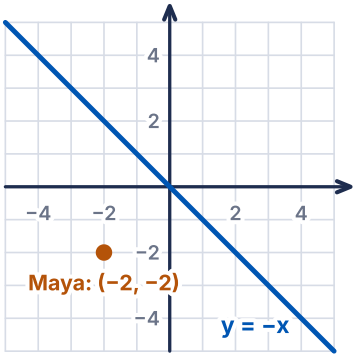
How to do it: y = −(−6) = 6. The opposite of −6 is 6.



3. ★★★ Challenge Maya says the point (−2, −2) is on the line y = −x. Is she right?

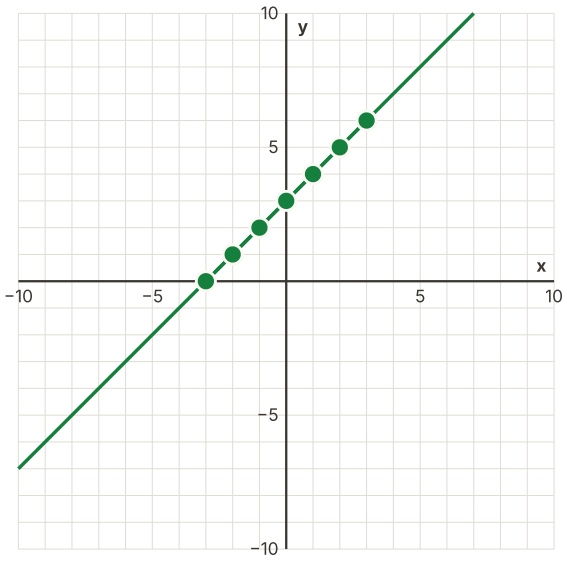
- ○ Yes, she is right
- ● **No. When x = −2, y = 2**
- ○ No. When x = −2, y = 0

How to do it: y = −(−2) = 2. So the point on the line is (−2, 2), not (−2, −2). Maya forgot to take the opposite.



3 y = x + 3

Input (x)	y = x + 3
−3	0
−2	1
−1	2
0	3
1	4
2	5
3	6



✏ Practice: y = x + 3

1. ★ Try it For y = x + 3, what is y when x = 2?

Answer: 5

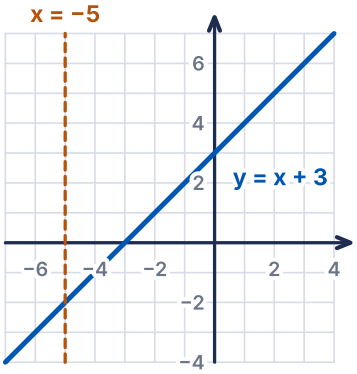
How to do it: y = 2 + 3 = 5.

x	y = x + 3
0	3
1	4
2	5

2. ★★ On level For y = x + 3, what is y when x = −5?

Answer: −2

How to do it: y = −5 + 3 = −2.



3. ★★★ Challenge On the line y = x + 3, what x-value gives y = 10?

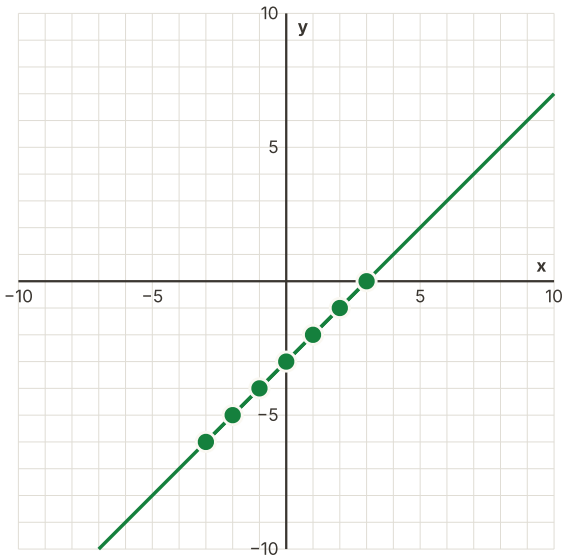
Answer: 7

How to do it: 10 = x + 3. Subtract 3 from both sides: x = 7. Check: 7 + 3 = 10 ✓

x	y = x + 3
0	3
1	4
7	10

4 y = x − 3

Input (x)	y = x − 3
−3	−6
−2	−5
−1	−4
0	−3
1	−2
2	−1
3	0



✏ Practice: y = x − 3

1. ★ Try it For y = x − 3, what is y when x = 5?

Answer: 2

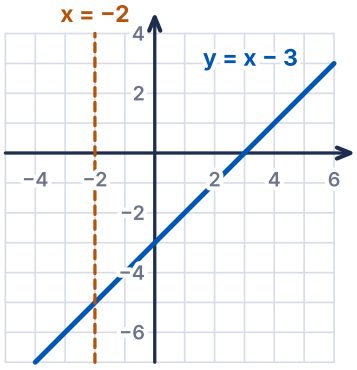
How to do it: y = 5 − 3 = 2.

x	y = x − 3
0	−3
1	−2
5	2

2. ★★ On level For y = x − 3, what is y when x = −2?

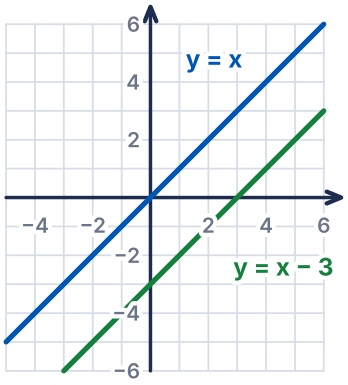
Answer: −5

How to do it: y = −2 − 3 = −5.



3. ★★★ Challenge Compare the graphs of y = x and y = x − 3. How is y = x − 3 different?

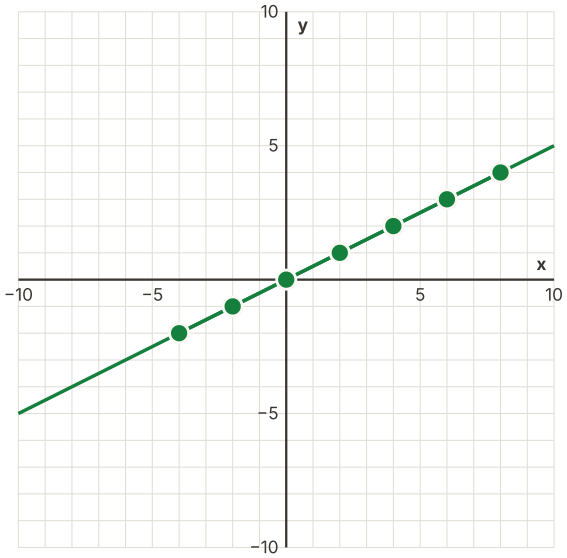
- ○ It is moved up 3
- ● **It is moved down 3**
- ○ It is steeper
- ○ It goes down instead of up



How to do it: Every y-value is 3 less than on y = x, so the whole line moves down 3. It crosses the y-axis at −3 instead of 0.

5 y = ½x

Input (x)	y = ½x
−4	−2
−2	−1
0	0
2	1
4	2
6	3
8	4



✏ Practice: y = ½x

1. ★ Try it For y = ½x, what is y when x = 10?

Answer: 5

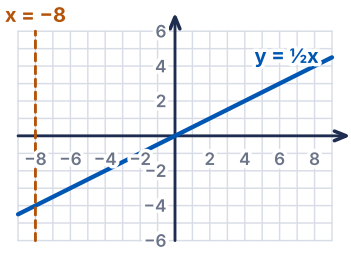
How to do it: y = ½ · 10 = 5.

x	y = ½x
0	0
2	1
4	2
10	5

2. ★★ On level For y = ½x, what is y when x = −8?

Answer: −4

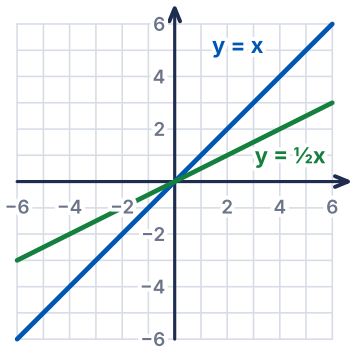
How to do it: y = ½ · (−8) = −4.



3. ★★★ Challenge Which line is steeper: y = x or y = ½x?

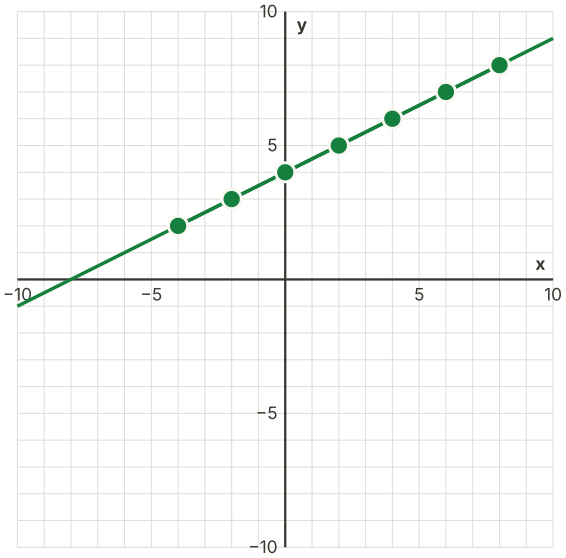
- ● **y = x**
- ○ y = ½x
- ○ They are equally steep

How to do it: On y = x, y goes up 1 each time x goes up 1. On y = ½x, y only goes up ½. So y = x is steeper.



6 $y = \frac{1}{2}x + 4$

Input (x)	$y = \frac{1}{2}x + 4$
−4	2
−2	3
0	4
2	5
4	6
6	7
8	8



✏ Practice: y = ½x + 4

1. ★ Try it For y = ½x + 4, what is y when x = 2?

Answer: 5

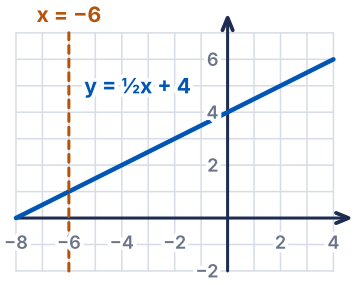
How to do it: y = ½ · 2 + 4 = 1 + 4 = 5.

x	y = ½x + 4
−2	3
0	4
2	**5**

2. ★★ On level For y = ½x + 4, what is y when x = −6?

Answer: 1

How to do it: y = ½ · (−6) + 4 = −3 + 4 = 1.



3. ★★★ Challenge A plant's height after x weeks is y = ½x + 4 (in cm). How tall is it after 10 weeks? Write just the number.

Answer: 9

How to do it: y = ½ · 10 + 4 = 5 + 4 = 9. The plant is 9 cm tall.

Weeks (x)	Height in cm (y)
0	4
2	5
4	6
10	**9**

Reflection Questions

A. Compare #1 and #2. What effect does the negative sign have on the graph?

The negative sign flips the line. y = x goes up from left to right, but y = −x goes down from left to right.

B. Compare #1, #3, and #4. What effect does adding or subtracting a number have?

Adding 3 moves the line up 3. Subtracting 3 moves it down 3. It changes where the line crosses the y-axis, but not how steep it is.

C. Compare #1 and #5. What effect does changing the number in front of x have?

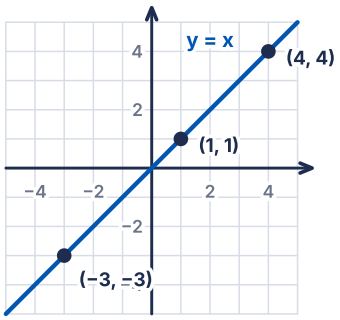
It changes how steep the line is. With ½ in front of x, the line is less steep: y goes up 1 for every 2 steps to the right.

✏ Comprehensive Practice All sections

1. ★ Try it On the line y = x, the x-value and the y-value are always...

- ● **equal**
- ○ opposites
- ○ 3 apart

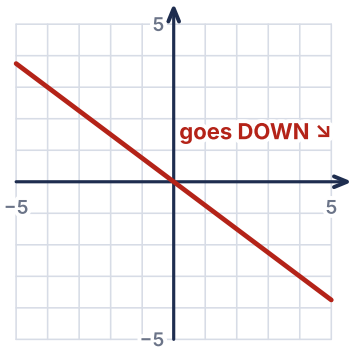
How to do it: y = x means y is the same number as x, so they are always equal. Example: (4, 4).



2. ★ Try it Which line goes DOWN from left to right?

- ○ y = x
- ● **y = −x**
- ○ y = x + 3
- ○ y = ½x

How to do it: The negative sign in front of x flips the line. So y = −x goes down as you move to the right.



3. ★★★ Challenge Jordan says that on y = −x, when x = −3, y = −3. What is the correct y?

Answer: 3

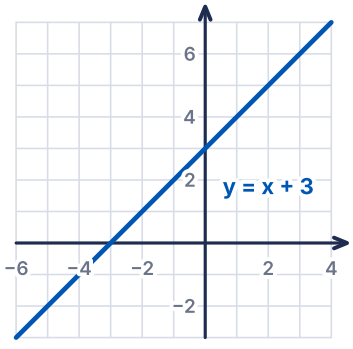
How to do it: y = −(−3) = 3. Jordan forgot that the opposite of a negative number is positive.

x	y = −x
−1	1
−2	2
−3	**3**

4. ★ Try it Where does the line y = x + 3 cross the y-axis? Write the y-value.

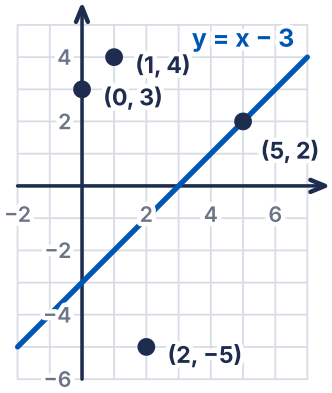
Answer: 3

How to do it: On the y-axis, x = 0. So y = 0 + 3 = 3. The line crosses at (0, 3).



5. ★★ On level Which point is on the line y = x − 3?

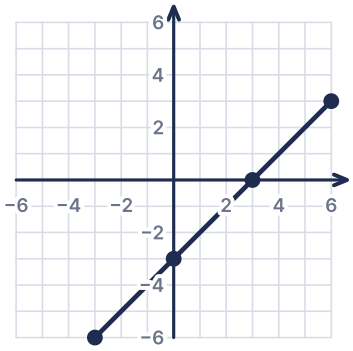
- ○ (1, 4)
- ● **(5, 2)**
- ○ (0, 3)
- ○ (2, −5)



How to do it: 5 − 3 = 2 ✓ so (5, 2) works. The others don't: 1 − 3 = −2, 0 − 3 = −3, 2 − 3 = −1.

6. ★★★ Challenge Which equation matches this graph?

- ○ y = x + 3
- ● **y = x − 3**
- ○ y = −x − 3
- ○ y = ½x − 3



How to do it: The line crosses the y-axis at −3, so the equation ends in − 3. It goes up 1 for every 1 step right, like y = x. So it is y = x − 3.

7. ★★ On level For y = ½x, what is y when x = 7? Write a decimal or a fraction.

Answer: 3.5

x	y = ½x
2	1
4	2
6	3
7	3.5

How to do it: y = ½ · 7 = 3.5 (or $\frac{7}{2}$).

8. ★★★ Challenge Which equation matches this table?

x	y
0	4
2	5
4	6
6	7

- ○ y = x + 4
- ● **y = ½x + 4**
- ○ y = ½x
- ○ y = 2x + 4

How to do it: When x = 0, y = 4, so the equation ends in + 4. When x goes up 2, y goes up 1. That is ½x. Check x = 6: ½ · 6 + 4 = 7 ✓

9. ★★ On level For y = −x + 2, what is y when x = 5?

Answer: −3

How to do it: y = −(5) + 2 = −5 + 2 = −3.

